TCS CORPORATE WEBSITE

Troubleshooting Runbook

Role / audience	Operations and support — AMS / support teams, site administrators
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Troubleshooting Runbook

Attribute	Value
Document ID	OPS-002
Audience	AMS / support teams, site administrators
Purpose	Triage, runbook and evidence
Scope	—
Prerequisites	—
Owner	AMS / Support Lead (proposed)
Approver	Product Owner (to be confirmed)
Version	1.0
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Source files	—
Related documents	—
Related AEM paths	—

1. Triage First

1. **Where does it happen?** Universal Editor (author content, `main` code), preview (`*.aem.page`), live (`*.aem.live`) or local (`localhost:3000`, local code on previewed content). This separates authoring state, publication state, caching and code.
2. **Which locale?** Most site data is per locale (nav, footer, placeholders, taxonomy, sheets, 404).
3. **How wide?** One page points to content; one block type to block code or its content; every page to code, configuration or the platform.
4. **What changed just before?** A merge, a publication, a package install or a configuration change.
5. **Which breakpoint?** Mobile < 600px, tablet 600–1199px, desktop ≥ 1200px (the header's desktop layout also starts at 1200px).

Diagnostic toolkit

Tool	Use
Console	`failed to load module for <block>`, `failed to load block <block>`, `failed to load template <name>`, `Auto Blocking failed`, `failed to play the Lottie animation`, `motion` / `skeleton` errors, CSP violations
Network	Block JS / CSS, `*.plain.html` (fragments, nav, footer, error pages), `/{locale}/placeholders.json`, `/{locale}/taxonomy.json`, `/{locale}/data/*.json`, `/query-index.json?offset=0&limit=500`
Delivered markup	`curl https://main--tcs-corp--sradha-tcs.aem.page/{path}.plain.html` shows the rows and cells a block receives; `curl -I` shows status and cache headers
Debug switches	`?debug=list`, `?debug=separator` (`console.debug`, enable Verbose); `?rum=on` forces RUM sampling
Data layer	`window.adobeDataLayer`, `adobeDataLayer.filter((e) => e.event)`
Journey state	`document.documentElement.dataset.canvasJourneyState` (Conversation Canvas)

Tool	Use
Admin API	`GET https://admin.hlx.page/status/sradha-tcs/tcs-corp/main/{path}` (preview, live, source times; authenticated)
Local server	`aem up` to tell code from content

2. Runbook

Development and code

Symptom	Likely cause	Diagnostic steps	Resolution	Verification
Local dev server doesn't start or pages 404 on `localhost:3000`	AEM CLI not installed or the port is taken. The page isn't previewed: the local server serves previewed content. Node older than 20 (README: Node 20+; CI uses 24)	`node -v`; read the CLI output; open the page on `main--tcs-corp--sradha-tcs.aem.page`. For test pages, check the file is in `drafts/`	`npx -y @adobe/aem-cli up --no-open --forward-browser-logs`; free the port; preview the page in AEM, or serve static pages with `--html-folder drafts`	The page loads on `localhost:3000` with the local code
Pre-commit doesn't rebuild `component-*.json`	`npm install` never ran (Husky is installed by `prepare`), or the commit used `--no-verify`	Check `.husky/pre-commit` runs; `npm run build:json` then `git diff`	`npm install`; `npm run build:json` and commit the three aggregated files	No diff after `npm run build:json`
`npm test` fails in `module-graph` or `block-docs`	An import or model changed without regenerating	Read the failing assertion	`npm run build:modules` and / or `npm run build:docs`; commit the output	`npm test` passes
Build check red on a PR	ESLint (Airbnb, `.js` extensions, line endings), xwalk model rules or Stylelint	Actions log; `npm run lint` locally	`npm run lint:fix`; for `xwalk/max-cells` use field collapsing / element grouping before raising the limit in `.eslintrc.js`	Build green on the latest commit
Block shows raw rows or nothing; `failed to load module for <block>`	JS / CSS 404, syntax error, `decorate()` threw on unexpected content, block not on `main` yet, or a renamed import	Console stack; Network status of `/blocks/{name}/{name}.js`; `.plain.html` of the block vs its `_{name}.json`; reproduce with `aem up`	Fix the code (or revert the merge), or fix the content; merge the branch that adds the block	No load error in the console. Note: `data-block-status="loaded"` is set **even when loading failed**, so it doesn't prove success
Decoration error only on the published page, not in the editor or local preview	The code assumes a structure AEM doesn't deliver: an empty row for each dialog tab, item first cell `<node name>`, `<classes>`, collapsed `*Text` / `*Title` / `*Type` / `*Alt` fields, lower-cased item names	Compare `.plain.html` on aem.page with the local markup; count the item's cells	Read rows by `data-aue-prop` in the editor and skip tab rows on the page (`itemRows`, `utils/cells.js`); never assume "the first row"	The block renders on aem.page; unit tests for the cell layout pass
Block CSS not applied although the CSS file exists	`loadCSS` (`scripts/aem.js`) skips a stylesheet when a `head > link` with the same href exists, **whatever its rel**: a `preload` link added to the head suppresses the stylesheet. Other causes: CSS 404; selector not scoped or overridden; style only in `lazy-styles.css`	Elements panel: is there a `rel="stylesheet"` link for `/blocks/{name}/{name}.css`? Is there a head link with the same href? Network status	Put stylesheet preloads in the body (as `preload` in `scripts.js` does); fix the path or the selector	Computed styles show the block rules; no 404

Symptom	Likely cause	Diagnostic steps	Resolution	Verification
Universal Editor reloads the page on every edit, or shows stale rendering	`editor-support.js` couldn't match the update to an element and called `window.location.reload()`. Container sections re-render as a whole. Blocks that read the index or sheets can't show published data	Console during the update; which block	Reload the editor; make the block re-decoratable from fresh markup; report with the block and page path	Edits of simple blocks apply in place
Code change not visible after merge	Code Sync didn't process the push; browser cache; the file is ignored by `.hlxignore` (`_*`, `docs/*`, `test/*`, `*.md`)	Code Sync check on the merge commit; request the file on `main--tcs-corp--sradha-tcs.aem.page`	Re-trigger the sync (Admin API code operation or an empty commit); hard reload	The served file matches the commit
Third-party script blocked (CSP)	`head.html` / `404.html` CSP `'nonce-aem' 'strict-dynamic'`: tags without the nonce are blocked	Console CSP message (blocked URI, directive)	Load the script with `loadScript()` from `scripts/delayed.js`; change the policy only in both files with a security review	No CSP violation; the script runs

Authoring and delivery

Symptom	Likely cause	Diagnostic steps	Resolution	Verification
Dialog field, tab or block missing in Universal Editor	Aggregated `component-*.json` not rebuilt; change only on a branch (the editor uses `main`); block not in the section or container filter; editor not reloaded; field hidden by its visibility condition	Search `component-models.json` on `main`; `npm run build:json` and diff; check `component-filters.json`; read the block README's field table (conditions)	Rebuild, merge, reload the editor; add the block to the filter; set the controlling field	The field or block appears after a full reload
A dialog value doesn't persist or changes nothing on the page	Delivery rules: the field name ends in Text / Title / Type / Alt / MimeType and collapses; grouped `prefix_name` field in a key-value block; `classes_*` on a section (not delivered; use `container*`); `multi: true` on `aem-content` (not saved); built-in Button / Columns options (not delivered)	`.plain.html` on aem.page: is the value there? For sections, look for `data-container*` attributes	Rename the field (`buttonVariant`, `titleHidden`, `poster`), use item children or `container*` fields; migrate content; use the Button Block / Columns Container	The value appears in the delivered markup and the block reflects it
Hidden block, or a faded block with a dashed outline	Additional tab Hide block (`hideblock`): hidden outside edit mode, dimmed in edit mode, still in the HTML	Select the block; check Hide block	Clear Hide block and republish (never use it for confidential content)	Visible in Preview and on the site
`#anchor` link doesn't jump	Anchor ID includes its `anchor-` prefix in the id; duplicates get `-2`; sections use Anchor name	Inspect the element `id`	Link `#anchor-…`; use unique ids	`page#anchor-…` scrolls to the block
Fragment, Modal or Tab content empty	Fragment page not published; path not starting with `/` (full URL) or `//`; the fragment references the page itself or its including chain (recursion guard)	Network: `{path}.plain.html` status; the Reference value	Publish the fragment; pick it with the content picker	Content renders on reload

Layout, media and links

Symptom	Likely cause	Diagnostic steps	Resolution	Verification
Layout wrong at one width, or horizontal scrolling	Rule outside the mobile-first breakpoints (600 / 900 / 1200px); fixed px sizes; an element wider than the viewport; the editor canvas is narrower than the window (e.g. Teaser image Right applies from 600px)	Reproduce at 1440, 1024, 768, 600 and 390px; in the console compare `document.documentElement.scrollWidth` with `clientWidth`; find the element extending past the right edge	Fix the block CSS (rem, `min-width` queries, scoped selectors); test the editor at its canvas width	No overflow at the five widths; the harness overflow check is 0
Header layout unexpected between 900 and 1199px	The desktop header (logo bar) starts at 1200px (`header.js` `matchMedia('(min-width: 1200px)')`); some handover pages still say 900px	Check the viewport width	Expected behaviour; test at ≥ 1200px for desktop	Logo bar at ≥ 1200px, menu below
Lottie, video or captions 404 on aem.page / aem.live while images work	Non-image DAM files (Lottie JSON, MP4, WebVTT) are served only once previewed / published; images travel with the page	Network: the asset URL status	Publish the assets from AEM Assets (publish with references)	The file returns 200; the animation or video plays
Lottie shows a still frame, or video waits for a click	Reduced motion (by design: paused animations, click-to-play videos); `failed to play the Lottie animation`; Image type not Lotto	OS reduced-motion setting; console; asset status	Accept reduced motion; publish the JSON; set the Image type	Plays with reduced motion off
Image not optimised or large	Typed / external URLs (only same-origin `/media_` images are optimised by `createOptimizedPicture` / `decorators/responsive-image-decorator.js`); large committed SVGs or Lottie files	Network (Img): `/media_…?width=…&format=webply` or external	Use AEM Assets references; compress committed files	`<picture>` with width variants; PageSpeed no longer flags it
Broken image in a fragment or card	Asset unpublished, not a `/media_` path, or a typed URL	Inspect `img src`	Re-select the image with the picker and republish	Image loads
Link opens wrong target or isn't announced	List cards link to the redirect target of `/{locale}/redirects.json` (`utils/redirects.js`, up to 5 hops); a new-tab link without `announceNewTab`; unsafe schemes are rejected (`isSafeUrl`)	Check the sheet; inspect the link's text / `aria-label`	Correct the redirects sheet; use the link decorators	Card URL as expected; new-tab hint present
Old URL returns 404 instead of redirecting; `/` returns 404	The platform applies only a root `/redirects.json`; per-locale sheets aren't applied; `/content/tcs-corp` isn't published	`curl -I` the old URL / the root	Configure the redirect mechanism (e.g. CDN rules); decision owned by the site administrator	301 / 302 with the expected `location`

Index, lists and search

Symptom	Likely cause	Diagnostic steps	Resolution	Verification
Page missing from `/query-index.json`	Not published to live; under an excluded path (`/*/references/**`, `/*/error-pages/**`, `/*/data/**`, `/*/nav`, `/*/footer`, block libraries, `site-global-templates`, `/**/fragments/**`, `/**/modals/**`); not reindexed after a `helix-query.yaml` change	Search the path in `/query-index.json?limit=500` (and further `offset` pages); check the excludes	Publish; move the page out of excluded folders; reindex	The row exists with the expected columns

Symptom	Likely cause	Diagnostic steps	Resolution	Verification
Index row lacks columns (card, eyebrow, hide-in-nav, featured)	The index was built before the `helix-query.yaml` change (finding F-10)	Inspect the row's keys	Reindex the affected paths	Columns present
List / Filter List empty in Universal Editor	Expected: no query index on AEM author; the hint says the pages show on the published page, or names missing settings	Read the hint	Complete the named settings; review on aem.page	Cards on the preview
List / Filter List incomplete on the site	Pages not published or indexed; Parent page / depth, Search root, tags (any / all), Root path, Page types / Content types don't match; `hide-in-nav = yes`; Maximum items; rows filtered to the locale root (`indexRoot`)	`?debug=list` (configuration, warnings, cards); compare index rows with the settings	Fix settings or metadata; publish the pages	Expected cards and result count
Filter List "No resources found." or pills with derived labels	Filters too narrow; tags missing from `/{locale}/taxonomy.json`; values that aren't tag IDs	Index rows below the Root path; the taxonomy JSON	Fix Root path / types; publish the taxonomy; pick tags with the picker	Cards and taxonomy titles
Canvas search does nothing visible	No Search page and no integration listening to `conversation-canvas:search`	Check the Search page field; listen for the event in the console	Set the Search page or connect an integration	A query opens results
Search term missing or altered in the data layer	By design `cmp:search` carries the redacted term (e-mail addresses and phone numbers replaced, cut at 100 characters; `utils/privacy.js`); an empty search pushes nothing	`adobeDataLayer.filter((e) => e.event === 'cmp:search')`	None (privacy rule)	Redacted term present

Data layer, analytics and Adobe integrations

Symptom	Likely cause	Diagnostic steps	Resolution	Verification
`adobeDataLayer.getState` / `addEventListener` undefined	Expected: the ACDL library isn't loaded (normally by the Tags ACDL extension), so `adobeDataLayer` is a plain array	`Array.isArray(window.adobeDataLayer)`	None until Adobe Tags is configured; read the array	Entries present in the array
Block or component entries missing	Blocks are added once decorated (main content in the lazy phase; header, footer and late blocks after 3 seconds in `loadDelayed`); `data-analytics="off"` on the block; blocks inside `[data-analytics]` push their own state	Wait 3 s; check `data-cmp-id`, `data-analytics` on the element	Turn the block's Analytics option on (List, Cards, Separator); report missing entries for tracked blocks	Component entry with `@type tcs-corp/components/{block}`
Clicks missing	Clicks on `<summary>` push `cmp:show` / `cmp:hide`, not clicks; the element is inside `data-analytics="off"`; a handler stopped the event before the capture listener	`adobeDataLayer.filter((e) => e.event === 'cmp:click')` with Preserve log on	Describe the interaction with `data-cmp-*`; don't stop propagation in capture	One `cmp:click` per click
Clicks duplicated, or duplicate component ids	A block pushes its own `cmp:click` in addition to the central capture; ids reused (`data-cmp-id` from duplicate block ids)	Count events per click; list component ids	Remove block-level click pushes (blocks only push state via `pushComponentEvent`); keep ids unique (Anchor IDs are de-duplicated with `-2`)	Harness click test: exactly one `cmp:click`; no duplicate ids

Symptom	Likely cause	Diagnostic steps	Resolution	Verification
No data in Adobe Analytics / CJA	Adobe Tags (Data Collection) and the Web SDK are **not configured**; `scripts/delayed.js` is empty	Check `delayed.js`; no Tags / `alloy` requests in Network	Blocked: Tags property, datastream and the measurement plan (handover 11.7)	After integration: AEP Debugger shows `sendEvent` calls
Target / AJO activity not shown	Not implemented: no Target, AJO or audience integration exists	—	Not applicable until a personalization design is approved	—
Consent banner missing, data before consent, identity questions	Not implemented: no consent mechanism or identity service; nothing is sent to third parties; the data layer stays in the page	—	Blocked: the consent model and CMP	—

Accessibility and performance

Symptom	Likely cause	Diagnostic steps	Resolution	Verification
axe contrast violations on a dark Section / Columns Container	Block uses colour primitives instead of the theme tokens; a block with a fixed light colour inside a dark background	axe at 1440 and 390px; the "Sections: colours with blocks" library page	Use the tokens (`--color-text-default`, `--link-color`, `--color-border-brand`, ...); reset tokens inside Grey islands	0 axe violations on the combination page
Keyboard trap, focus lost, Escape doesn't close	Disclosure without `<button>` / `aria-expanded` / `aria-controls`; focus not returned on close	Keyboard pass (Tab, Shift+Tab, Enter, Space, Escape, arrows)	Follow AGENTS.md "Accessibility"; return focus to the opener	Keyboard-only pass succeeds
English UI labels on another locale	Placeholder key missing or the sheet unpublished; built-in `404.html` texts aren't placeholders	`/{locale}/placeholders.json`; find `t('key', 'fallback')`	Add the key to the locale sheet and publish	Translated label on reload
LCP, TBT or CLS regression	Stale preload lists; forced layout during decoration; heavy or lazy-image first section; scripts outside `delayed.js`; fallback font faces removed; content arriving without reserved space	PageSpeed / Lighthouse diagnostics; Performance panel; `npm test` (module graph)	`npm run build:modules`; batch reads / writes in one animation frame; keep the LCP image eager in the first block; keep `inter-fallback*` faces	PageSpeed back to target; RUM CLS ≤ 0.1, LCP < 2.5 s

Packages, publishing and reindexing

Symptom	Likely cause	Diagnostic steps	Resolution	Verification
After installing a content package, pages don't appear on aem.page / aem.live	Installing in AEM doesn't preview or publish; sheets (placeholders, metadata) are separate	Admin API status per path; Sites console publication status	Publish sheets first, then fragments, then pages (handover 12.4); or Admin API `preview` then `live` per path; retry paths that time out (one did in 1.0.12)	Every page previewed and published (e.g. 315 / 315 for 1.0.13)
Packaged pages render without CTAs, stats, filters or footer links	Content written as composite multifields (`ctas/item0`) doesn't render (finding F-87)	`.plain.html` : empty item rows	Use item children (`hero-cta`, `teaser-link`, `footer-link`, ...)	Items render on the published page

Symptom	Likely cause	Diagnostic steps	Resolution	Verification
Package procedure itself (build, install) unclear	Not recorded in the repository: the packages are built in the delivery workspace	—	Blocked: package build and install procedure from the delivery team	—
New pages missing from lists after publishing	Not yet indexed, or the index predates a `helix-query.yaml` change	Search `/query-index.json`	Reindex: POST `https://admin.hlx.page/index/sradha-tcs/tcs-corp/main/{path}` per path (authenticated); publish sheets before pages so rows carry their metadata	Row present (e.g. 312 / 312 reindexed for 1.0.10); block library pages stay out by design
Live page still old after publish	Published to preview only; a reference (fragment, nav, footer, asset, sheet) not published; caching; inherited Navigation / Footer / 404 values cached per session (`sessionStorage` `tcs-inherited-metadata`)	Compare aem.page / aem.live in a private window; Admin API status	Republish with references; purge the cache; new browser session	Fresh `last-modified`; change visible

3. Further Site-Wide Issues

Each of these has symptoms, root cause, diagnosis, resolution, validation and escalation in the handover's Part 13 — Global Troubleshooting Guide (TCS-CORP-HANDOVER.md):

Issue	Handover section
Pages published at the wrong path, locale missing	13.4 (`paths.json` and `public.json` paths must match)
Wrong header or footer	13.6 (inherited Navigation / Footer, `<meta name="nav">`)
404 in the wrong locale or the built-in English 404	13.7 (`loadErrorPage` fallback chain)
Tags showing IDs	13.14 (`/{locale}/taxonomy.json`)
Bulk metadata not applied	13.17 (site configuration `metadata.source`)
Form submit sends nothing	13.21 (`form-variations:submit`; see also the Form Variations README)
Embeds not loading	13.24
Sitemap host	13.25 (`helix-sitemap.yaml` `origin`)
Template (Page Type) not applied	13.26 (TEMPLATES in `scripts.js`: home, article, blog, press-release)
Page Header author details missing	13.27 (Authors sheet)

Block-specific runbooks: the "Troubleshooting" section of each block README, for example List, Filter List, Header, Fragment, Video, Hero, Teaser, Form Variations, Conversation Canvas.

4. Evidence to Collect

Collect:

- the page path (AEM and published) and the environment (UE / aem.page / aem.live / local);
- the locale, viewport width, browser and OS (including the reduced-motion setting);
- the time in UTC and the related merge, publication or package;
- console messages with stacks;
- Network status codes of the block, sheet, fragment and index requests;
- the block's `.plain.html` excerpt and outer HTML;

- the block's dialog settings (screenshot), the `?debug=list` / `?debug=separator` output, the relevant `/query-index.json` rows;
- `copy(JSON.stringify(adobeDataLayer, null, 2))` with `mailto:` / `tel:` link URLs removed.

Don't collect:

- cookies, tokens or Authorization headers;
- author host names;
- AEM program or environment ids;
- form values;
- visitors' personal data;
- unredacted search terms;
- screenshots showing personal details.

5. Related

- development-standards.md, testing-and-quality.md, universal-editor.md, decorators-and-utilities.md, architecture.md, indexing-and-search.md, data-layer-and-analytics.md, security-privacy-consent.md, personalization.md, administration-and-operations.md, deployment-and-release.md
- Handover (TCS-CORP-HANDOVER.md): Part 13 — Global Troubleshooting Guide, 14.3 Logs, 14.4 Common Errors, 14.5 Escalation Matrix, 12.6 Rollback
- AGENTS.md "AEM delivery rules", "Module preloading"; blocks/README.md; docs/audit/REAUDIT-2026-10-20.md (operations and content actions)

6. Screenshots

No screenshot belongs to this document (it describes code, configuration or process). Evidence screenshots are listed in the screenshot register.

7. Known limitations

- None recorded for this document.

8. Open items

- None.

9. Change history

Version	Date	Author	Change
1.0	8 November 2026	Delivery team (documentation program)	First version of the role-specific documentation (master documentation prompt), generated from the repository and the published reference pages